Standardized monotonic and diverging colormaps

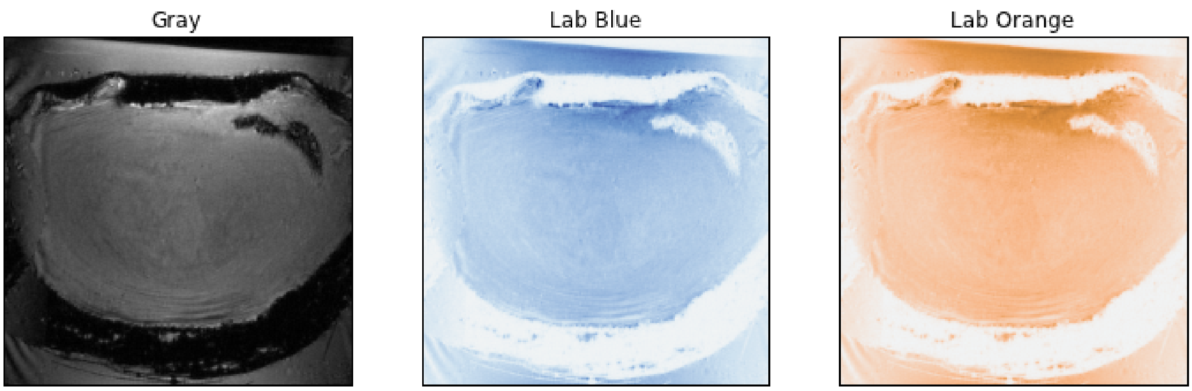
When preparing plots, we often use colormaps to make the information more accessible. There are two major use cases: monotonic and diverging data. Monotonic data ranges from a minimum to a maximum, with no natural break in between. Diverging data does have a natural break, for instance values on the range $[-a, a]$. A third use case is for categorical data, which will not be discussed. For monotonic data, the ideal is a single-tone colormap smoothly varying in intensity from light to dark. A grayscale colormap is the best choice from a technical perspective, but can make a publication look somewhat drab. Diverging data is ideally represented by a two-tone colormap centered on either white or black.

For convenience, several colormaps are provided here to meet these needs. The examples are prepared using python, but instructions are also provided to obtain equivalent results in Matlab. The `colormap` package for python can be downloaded from `128.175.10.99/peloquin/repos/common/colormap.git`.

Monotonic colormaps

The monotonic usage case is fairly easy to satisfy, as any single-hue intensity scale will suffice. The two monotonic colormaps illustrated here are intended to match the diverging colormap presented below. Note, however, that the grayscale colormap provides superior contrast.

In [144]:
```
import os, sys
import matplotlib.pylab as pylab
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import matplotlib.mlab as mlab
import numpy as np
import colormap # This is a custom package written by John Peloquin
img = mpimg.imread("discgs.png")
pylab.rcParams['figure.figsize'] = 12, 8
fig, axes = plt.subplots(nrows=1, ncols=3)
axes[0].imshow(img, cmap="gray")
axes[0].set_title("Gray")
axes[1].imshow(img, cmap="lab_monotonic_blue")
axes[1].set_title("Lab Blue")
axes[2].imshow(img, cmap="lab_monotonic_orange")
axes[2].set_title("Lab Orange")
for ax in axes.flat:
    ax.get_xaxis().set_ticks([])
    ax.get_yaxis().set_ticks([])
```



Another popular monotonic colormap is called "hot". It is a little more colorful than the single-tone colormaps, but still maintains a monotonic intensity profile.

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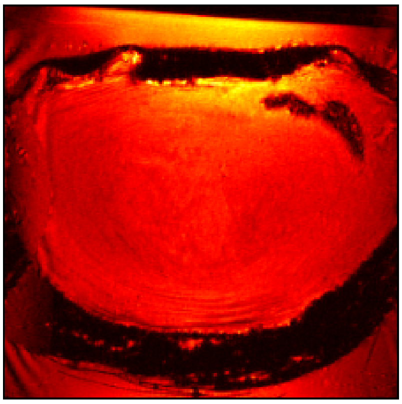
In [140]:
```
fig, ax = plt.subplots(figsize=(4, 4))
ax.imshow(img, cmap="hot")
ax.set_title("Hot")
ax.get_xaxis().set_ticks([])
ax.get_yaxis().set_ticks([])
```

Out[140]: []



Monotonic colormaps available in Matlab by default include:
`colormap('Gray')`
`colormap('Bone')`
`colormap('Copper')`
`colormap('Hot')`

Diverging colormaps

Data with both negative and positive values is best represented by a diverging two-tone colormap centered on zero. Here, the diverging usage case will be illustrated using a stress field around a crack.

This orange-blue colormap is not available in Matlab by default, but the lbmap package from the Matlab File Exchange includes it. Once `lbmap` is on the Matlab path, the colormap can be invoked in Matlab using `colormap(lbmap(256, 'brownblue'))`.

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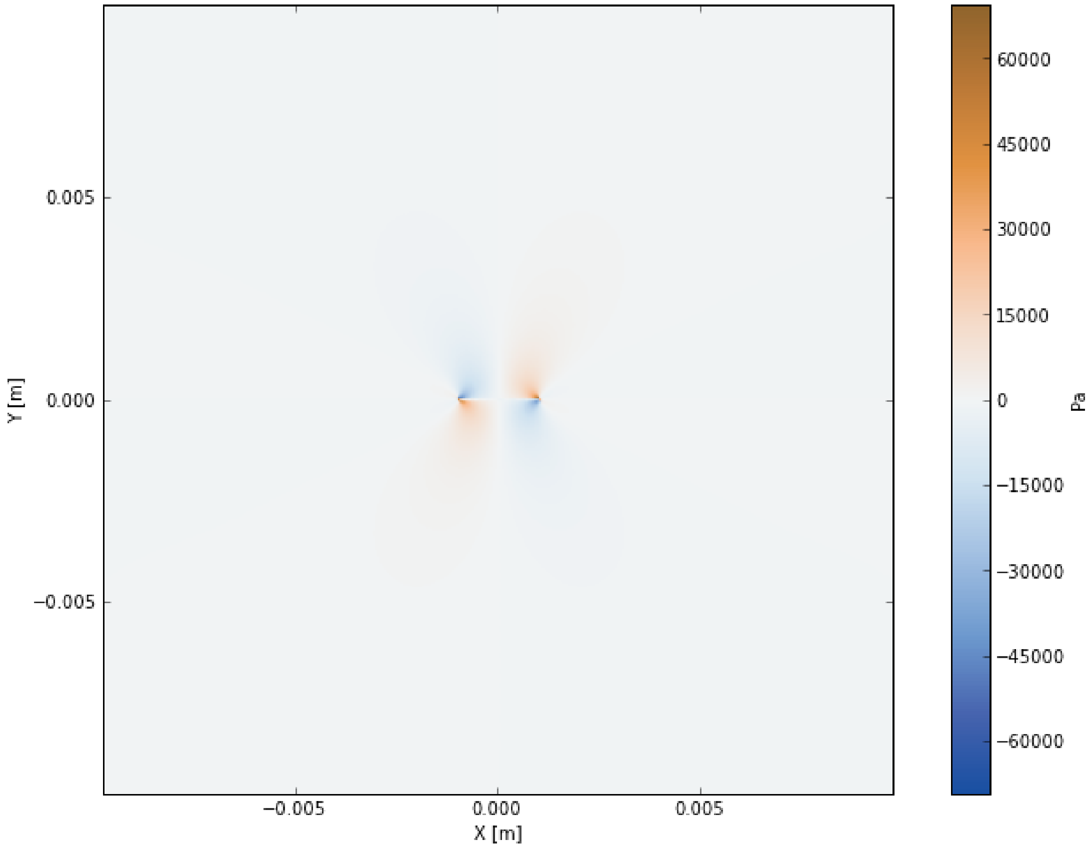
In [141]:
```
import febtools
soln = febtools.MeshSolution("center-crack-matrix-1mm-2d.xplt")
t, x, y, z = zip(*febtools.compare.midplane_stress(soln, type="cauchy"))
t12 = np.array([a[0,1] for a in t])
xi = np.arange(min(x), max(x), 50e-6)
yi = np.arange(min(y), max(y), 50e-6)
v = mlab.griddata(x, y, t12, xi, yi, interp='linear')
clim = max(abs(t12))
plt.imshow(v, extent=[min(x), max(x), min(y), max(y)],
                vmin=-clim, vmax=clim, cmap="lab_diverge")
plt.xlabel("X [m]")
plt.ylabel("Y [m]")
plt.colorbar().set_label("Pa")
```

```
Reading center-crack-matrix-1mm-2d.xplt
Found data named displacement in  node section, step 16.
Found data named stress in  domain section, step 16.
Found data named shell thickness in  domain section, step 16.
Found data named shell strain in  domain section, step 16.
```



Jet

Often, the "jet" colormap is used without regard for the type of data being presented (probably because it is very colorful). Unfortunately, jet is a spectral colormap, and produces artificially sharp intensity gradients in non-categorical data. The highest intensity (yellow) does not lie at the extreme of the data range, and the colormap is not interpretable by people with dichromacy or anomalous trichromacy.

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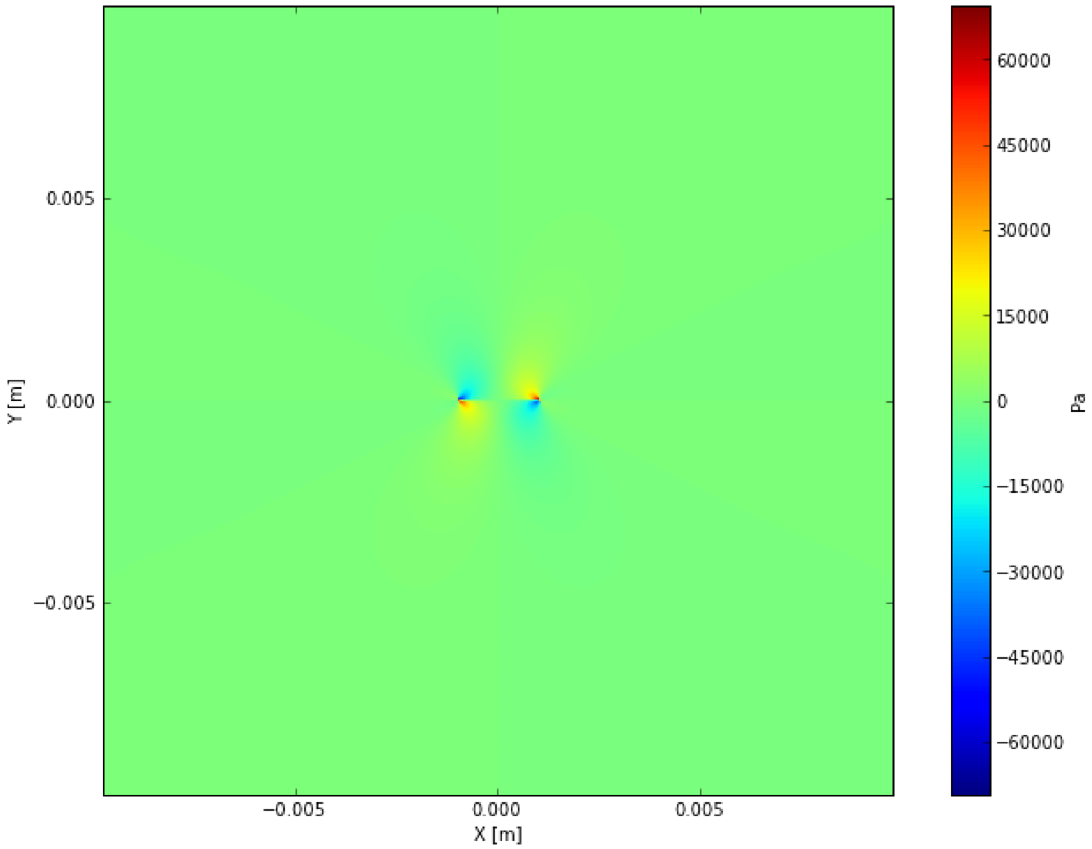
In [142]:
```
plt.imshow(v, extent=[min(x), max(x), min(y), max(y)],
                vmin=-clim, vmax=clim, cmap="jet")
plt.xlabel("X [m]")
plt.ylabel("Y [m]")
plt.colorbar().set_label("Pa")
```



Conclusion

Using appropriate single tone colormaps for monotonic data and two tone colormaps for diverging data facilitates correct interpretation by viewers. The examples provided here will hopefully facilitate adoption of best practices for display of colormapped data.

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